

Lecture 2 - Finite State Processes

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- FSPs are algebraic models used to represent the behaviour of sequential and concurrent processes.
 - FSPs are composed of **PROCESSES** and **actions**.
 - * An **action** is a thing which the system does.
 - An **action** must be atomic (can only do one thing).
 - * A **PROCESS** perform an action and calls another process.

Ex. Represent a process which performs action **x** and then calls another processes **P** as an FSP.

Solution:

$$(\mathbf{x} \rightarrow \mathbf{P})$$

- Repetitive behaviour is modelled by recursively calling processes.

Ex. Represent a simple on-off switch as an FSP.

Solution:

$$\begin{aligned}\mathbf{SWITCH} &= \mathbf{OFF} \\ \mathbf{OFF} &= (\mathbf{on} \rightarrow \mathbf{OFF}) \\ \mathbf{ON} &= (\mathbf{off} \rightarrow \mathbf{OFF})\end{aligned}$$

- *Process substitution* is the act of replacing processes with their definitions to produce more succinct FSPs

Ex. Using process substitution, use the FSP for Example 2 to product an FSP for an on-off switch with only two processes.

Solution:

$$\begin{aligned}\text{SWITCH} &= \text{OFF} \\ \text{OFF} &= (\text{on} \rightarrow (\text{off} \rightarrow \text{OFF}))\end{aligned}$$

- FSPs do not need to be deterministic.
- In FSP *choices* allow a process to execute processes non-deterministically.
 - Denoted as “|”.

Ex. Represent a process which can perform action x then execute process P or perform y then execute Q .

Solution:

$$(x \rightarrow P \mid y \rightarrow Q)$$

- A *boolean condition* is a statement which can either be true or false.
 - A boolean condition is said to have a *finite domain* if there exists a finite number of cases needed to be checked to evaluate it as true or false.
 - * For example, the boolean condition $x = 0$ has a finite domain, however $x > 0$ does not.
- A *guarded action* is an action in a choice can only be executed so long as some boolean condition is met.
 - The condition allowing a guarded action to be taken is specified with the **when** keyword.
 - The boolean condition must have a finite domain.

Ex. Represent a process which can perform action x then P ONLY IF a condition B is true or perform y then execute Q .

Solution:

$$(\text{when } B \ x \rightarrow P \mid y \rightarrow Q)$$

- The *alphabet* of a process is defined as the set of all possible actions that a process can perform.
 - The FSP representation of a representation implicitly defines the alphabet of a process.
 - If two FSPs have no actions in common, their alphabets are said to be *disjoint*.
 - An action is said to be *shared* between two processes if it exists in both of their alphabets.
- An *alphabet extension* is used to specify additional actions available to a process not defined in its FSP.
- If two processes are running concurrently, they are said to be running in *parallel composition*.
- The actions of a system composed of two concurrently running processes is an interleaving of the actions of each process.
 - The order of the actions taken by each system must be preserved.
 - * For example, if a process P can execute action **a** then **b**, and process Q can execute **c**, a valid output of the system (P||Q) could be $a \rightarrow c \rightarrow b$ but $b \rightarrow a \rightarrow c$ would be invalid, since in P, **a** must be executed before **b**.
- If two processes running in parallel composition have shared actions, then the shared actions must be executed simultaneously by both processes.
- FSPs use the parallel composition operator to declare two processes as running concurrently.
 - Denoted as “||”
 - The parallel composition operator is both associative and commutative.
- Petrinets allow for much easier graphical representations of concurrent systems than LTSs.